St. Philomena's College (Autonomous), Mysore
PG Department of Studies and Research in Chemistry
Question Bank (Revised Curriculum 2018 onwards)
Secomd year- Fourth Semester (2018 -20 Batch)
Bioinorganic Chemistry QP Code: 55301

Unit	SI. No	Questions	Marks
1	1.	What are nucleoside and nucleotide? Write its structure	2
1	2.	What are proteins? How are they classified based on their shape?	2
1	3.	What are essential and beneficial metals? What do you mean by essential trace elements?	2
1	4.	what is ATP? Write its structure.	2
1	5.	Which nitrogen bases are purines and which are pyrimidines and what is the difference between them?	2
1	6.	What types of interactions hold the secondary structure together?	2
1	7.	What types of interactions hold the tertiary structure together?	2
1	8.	Which of these amino acids are not optically active?	2
1	9.	How is the secondary structure of a protein stabilized?	2
1	10.	What do you mean by ionophores? Give examples of different types of naturally occurring ionophores acting as antibiotic.	2
1	11.	What is macro cyclic crown ether? Give an example.	2
1	12.	What are ionophores? Give examples for the naturally occurring ionophores and model compounds of naturally occurring ionophores?	2
1	13.	Identify the properties of crown ether-alkali metal complexes.	2
1	14.	Justify the names of crown ether, cryptate complex and football ligand.	2
1	15.	Categorize the ionophores in terms of mechanism of ion transport.	2
1	16.	What do you mean by channel forming ionophore and carrier ionophore?	2
1	17.	Crown ethers are not preorganised while spherands are preorganised for complexation- Justify.	2
1	18.	What do you mean by active transport and passive transport of ions?	2
1	19.	Validate on the role of Mg^{2+} in the activity of Na^{+}/K^{+} pump.	2

1	20.	Comment on the vanadate interference with the activity of Na^+/K^+ pump.	2
1	21.	Comment on the selectivity of Na^+/K^+ pump in transporting the Na^+ and K^+ ion.	2
1	22.	What is Na^+/K^+ pump? How does it function?	2
1	23.	What happens after the Na^+/K^+ pump is phosphorylated?	2
1	24.	What happens after Na^+/K^+ pump dephosphorylation?	2
1	25.	What is the purpose of pumping sodium and potassium across a membrane?	2
1	26.	Which are the binding sites of calcium in proteins?	2
1	27.	Which are the binding sites of calcium in blood clotting mechanism and in muscle contraction.	2
1	28.	What is the role of Vitamin K in blood clotting? Give examples for the antagonists of Vit K.	2
1	29.	Very often, Ca^{2+} protein interaction is stronger than Mg^{2+} protein interaction though Mg^{2+} possess a higher ionic potential. Comment.	2
1	30.	What is the distinction between the intrinsic and extrinsic pathway of blood clotting?	2
1	31.	Name the novel amino acids designed by nature for chelation of Ca^{2+} in the blood clotting process. How can you justify this nature's design?	2
1	32.	What are siderophores? Explain	2
1	33.	What are siderophores? Explain why high-spin Fe^{3+}complexes of siderophores are kinetically labile whereas analogous model complexes containing Cr^{3+} are kinetically inert.	2
1	34.	What is the function of transferrin and nitrogenase enzyme?	2
1	35.	What are the function of cytochrome and transferrrin?	2
1	36.	What is the biological function of ferritin and transferritin?	2
1	37.	What is oxidative phosphorylation? Why nature avoids direct oxidation of NADH by O_2.	2
1	38.	What is oxidative phosphorylation? Name the important electron carriers.	2
1	39.	Why nature avoids direct oxidation of NADH by O_2?	2

1	40.	What is ubiquinone? Contranst between plastoquinone and ubiquinone.	2
1	41.	Identify the ATP generating sites in the mammalian respiratory chain.	2
1	42.	Why does nature use so many electron carriers instead of direct oxidation of fuel molecules by oxygen?	2
1	43.	Report the acitivity of succinate-Q reductase in the respiratory chain.	2
1	44.	Report the activity of complex III in the mammalian repiratory chain.	2
1	45.	Which are the site III blocking agents in the respiratory chain? How do they arrest the electron flow?	2
1	46.	Name some important enzymes dependent on B_{12} coenzyme.	2
1	47.	Enumerate the characteristic features of B_{12} coenzyme.	2
1	48.	Indicate the functions of cyanocobalamin.	2
1	49.	How is corrin ring different from porphyrin ring?	2
1	50.	Give the names of different forms of cobalamins.	2
1	51.	Comment on the different oxidation states of cobalt in vitamin B_{12} to explain its reactivity.	2
1	52.	What is cobamide and cobaloxime?	2
1	53.	Give the characteristic properities of B_{12} coenzyme.	2
1	54.	What are the important biological functions of vitamin B_{12}.	2
2	55.	What are the basic requirements of a dioxygen complex to function as an oxygen carrier?	2
2	56.	Give examples of synthetic oxygen carriers.	2
2	57.	Comment on the thermodynamic and kinetic aspets of dioxygen as an oxidant.	2
2	58.	Spontaneous burning of biological molecules by oxygen is prevented because of the kinetic inertness of O_2 – Justify the statement.	2
2	59.	Explain the term Bohr effect in haemoglobin.	2
2	60.	What are the oxidation state and spin states of the metal ion in oxy-Hb and doxy-Hb?	2

2	61.	Explain the different oxygen binding behavior of the Co(II) prophyrin.	2
2	62.	How is Vaskas complex useful as synthetic oxygen carrier?	2
2	63.	Give examples of synthetic oxygen carriers.	2
2	64.	Give an example each for the synthetic oxygen carriers involving iron and cobalt.	2
2	65.	Differentiate between T and R structure of hemoglobin.	2
2	66.	Comment on the oxidation states of iron in the dioxygen complex of Hb and Mb	2
2	67.	What are hemocyanin and hemerthrin?	2
2	68.	Give examples of nonheme proteins involved in oxygen transport.	2
2	69.	Comment on the diamagnetic properties of oxy-Hr and oxy-Hc.	2
2	70.	Which are the metal ions present in Hc and Hr metalloproteins.?	2
2	71.	Write the structure of deoxyhemerythrin.	2
2	72.	Write the structure of doxyhemocyanin.	2
2	73.	The lower redox potential is observed in ruberodoxins, Give reason.	2
2	74.	What is Rieske protein? What is the difference between Rieske protein and Fe_2S_2(Fd)?	2
2	75.	What are the general features of iron sulfur proteins?	2
2	76.	What are iron sulfur proteins? Give examples of different types of ferredoxins.	2
2	77.	Would cytochrome-c be a useful oxygen carrier? Explain.	2
2	78.	What are cytochromes? Draw the structure of cytochrome c.	2
2	79.	How do you classify the cytochromes? Illustrate with examples.	2
2	80.	What are cytochrome? What is their function?	2
2	81.	What is the biological function of carboxypeptidase A?	2
2	82.	Give examples of the zinc containing metalloenzymes.	2
2	83.	What is the biological function of carbonic anhydrase(CA)?	2
2	84.	What is the biological function of CPA?	2

2	85.	What is the biological function of alcohol dehydrogenase?	2
2	86.	Which is metal ion present in the CPA and what is its coordination number?	2
2	87.	Sketch the structure of superoxide dismutase.	2
2	88.	Which are the metal ions present in superoxide dismutase?	2
2	89.	What are the functions of catalase and peroxidase?	2
2	90.	What is the biological function of superoxide dismutase? Mention different types of SOD.	2
2	91.	What is cytochrome P-450? How do you justify its name?	2
2	92.	Write the importance of sulfite oxidase.	2
2	93.	Give examples of molybdenum containing enzymes.	2
2	94.	Comment on the activity of xanthine oxidase.	2
2	95.	What is the function of nitrate reductase?	2
3	96.	What is the role of metals in medicine?	2
3	97.	How metals are important in human biochemistry.	2
3	98.	Give the example of metal Complex found in medical field.	2
3	99.	Write the structure of metal complex found in (a) Blood, (b) Plant pigment	2
3	100.	Write the structure of hemoglobin and chlorophyll.	2
3	101.	What is the role of iron in transport of oxygen in human body?	2
3	102.	What is the role of Mg metal in photosynthesis?	2
3	103.	Name the two diseases which cause internal bleeding and causes anemia.	2
3	104.	Mention the range of daily intake of zinc (Zn) in men and women.	2
3	105.	Name the Diseases due to the deficiency of zinc (Zn) in children.	2
3	106.	Name the micro mineral metal and trace metal required for healthy growth of body.	2
3	107.	What are the symptoms of the disease which cause due to the deficiency of copper in human body.	2

3	108.	What are the causes of selenium metal deficiency?	2
3	109.	Write the symptoms and treatment for deficiency of selenium metal in human body.	2
3	110.	What is the hyponatremia?	2
3	111.	What are the causes of hyponatremia?	2
3	112.	Write the symptoms of hyponatremia disease	2
3	113.	What are the causes of hypokalemia.	2
3	114.	Write the normal potassium level in blood and hence write its deficiency cause disease.	2
3	115.	What are symptoms of hypokalemia disease?	2
3	116.	Name the disease caused due to deficiency of fourth abundant metal or cation in the human body.	2
3	117.	What is hypomagnesaemia?	2
3	118.	Write the important role of magnesium in human body.	2
3	119.	What are the symptoms of hypomagnesaemia?	2
3	120.	How to treatment a person with hypomagnesaemia disease	2
3	121.	Name the diseases caused due to the deficiency of calcium in human body.	2
3	122.	What are the symptoms and treatment of hypocalcaemia disease?	2
3	123.	Write the structure of tetracycline and Ethambutol.	2
3	124.	Name the chelate used for the treatment agents a microorganisms like staphylococcus pyrogenes	2
3	125.	Write any structure of $[Fe(Me_4phen)_3]^{2+}$.	2
3	126.	Write the structure of Chelating agent and metal complex having antimicrobial activities.	2
3	127.	What is Arthritis? Name the metal complex used to treat rheumatoid arthritis.	2
3	128.	Write any two gold complex used in treatment of rheumatoid arthritis.	2

3	129.	Mention any four side effects of gold metal complex used in treatment of rheumatoid arthritis	2
3	130.	Write the two structure of metal complex used in cancer therapy.	2
3	131.	Mention any four properties of metal complex as anticancer agent.	2
3	132.	Cisplatin is used in cancer therapy not transplatin Give reason.	2
3	133.	What is Diabetes mellitus? Name the metal used to develop anti diabetes drug.	2
3	134.	Write any two structure of metal complex used as anti diabetic drug.	2
3	135.	Write any two vanadium compounds tasted as insulin mimetic agent.	2
3	136.	Write the structure of (i) BMOV (ii) $VO(acac)_2$	2
3	137.	Write the structure of (i) $VO(Me\text{-}acac)_2$ (ii) $VO(Et\text{-}acac)_2$	2
3	138.	Write the structure of (i) BMOV (ii) $VO(Et\text{-}acac)_2$	2
3	139.	Write any two side effect of vanadium Complex used as anti diabetes drug.	2
3	140.	What are the thermodynamically and kinetically possible oxidation state of vanadium metal and hence write the structure of BMOV.	2
3	141.	Compare the difference between hypoxia and histo hypoxia.	2
3	142.	Name the antidote used for a poison like mercury.	2
3	143.	Suggest any two antidote used in the treatment toxicity of metal ion.	2
1	144.	Explain the structure of proteins.	5
1	145.	Discuss briefly about the building blocks of nucleic acids.	5
1	146.	What are the characteristic features of primary & secondary structure of proteins?	5
1	147.	Discuss the probable mechanism of active transport of cations across the biological membrane through ionophores.	5
1	148.	Give the distribution pattern of ions in the cellular fluid. Give the thermodynamic aspects of the metal complexation with these macrocyclic ligands.	5

1	149.	discuss the properties of crown ether alkali metal complexes.	5
1	150.	What is valinomycin? Discuss its structural features. Explain the mechanism of antibiotic acitiviy of valinomycin.	5
1	151.	Discuss briefly on channel forming ionophore and carrier ionophore.	5
1	152.	Discuss the mechanism of K^+ transport by considering a model ionophore having a carboxylic acid group.	5
1	153.	What do you mean by eversion in the activity of Na^+-K^+ pump.	5
1	154.	What is Na^+-K^+ pump? How does it function? Comment on the role of Mg^{2+} in Na^+-K^+ pump.	5
1	155.	What do you mean by eversion in the activity of Na^+-K^+ pump. Comment on the selectivity of Na^+-K^+ pump in transporting the Na^+ and K^+ ions.	5
1	156.	Explain the intrinsic and extrinsic pathway of blood clotting mechanism.	5
1	157.	Describe the intrinsic pathway of blood clotting process.	5
1	158.	Explain the intrinsic and extrinsic pathway of blood clotting mechanism.	5
1	159.	Explain how Ca^{2+} ion makes conformational change after binding the active site of Troponin-C and explain how this alteration regulates muscle contraction.	5
1	160.	Discuss the role of Ca^{2+} in muscle contraction.	5
1	161.	Discuss the role of calcium in blood clotting process.	5
1	162.	Explain the mitochondrial flow of electrons from NADH to O_2.	5
1	163.	Draw the respiratory chain in terms of electron carriers. Why does nature avoids direct oxidation of NADH or $FADH^{2+}$ by oxygen?	5
1	164.	Identify the ATP generating sites in the mammalian respiratory chain.	5
1	165.	what are the different blocking agent in the respiratory chain?	5
1	166.	Explain the mitochondrial electron transfer chain. Mention the role of quinine in it.	5
1	167.	Discuss the structure and biological functions of ferritin.	5
1	168.	Explain the iron binding by transferrin	5
1	169.	Discuss the structure and biological functions of phosvitin.	5

1	170.	Give the coordination environment of cobalt in vitamin B_{12} coenzyme. Explain any two important biochemical reactions catalyzed by it.	5
1	171.	Discuss the structural features of vitamin B12. Give the names of different forms of cobalamins.	5
1	172.	Discuss the chemistry of cobalamin in different oxidation states of cobalt.	5
1	173.	what are the important biological functions of cobalamin?	5
2	174.	Discuss the general features of bonding mechanism in metal-dioxygen complexes.	5
2	175.	Discuss the different possible modes of binding of O_2.	5
2	176.	Explain with a schematic view the tetrameric structure of the hemoglobin.	5
2	177.	Discuss the structure of hemoglobin and its role in oxygen transport mechanism.	5
2	178.	Discuss the structural features of hemoglobin and myoglobin.	5
2	179.	Discuss the different interaction among the globin protein chains to provide the T-form and R-form of hemoglobin.	5
2	180.	Discuss the role of hemoglobin in maintaining the physiological pH and CO_2 transport from tissues to lungs.	5
2	181.	Compare the structure and reactivity of hemerythrin and hemocyanin?	5
2	182.	Explain O_2 binding in hemerythrin and hemocyanin.	5
2	183.	Discuss the structural features of hemerythrin and hemocyanin and their role in oxygen transport.	5
2	184.	Compare the fate of O_2 during oxygenation of hemoglobin, hemerythrin and hemocyanin.	5
2	185.	Write a note on iron sulphur proteins.	5
2	186.	What are ferridoxins and ruberodoxins? How do they differ structurally?	5
2	187.	What is Ruberodoxin? Discuss its structural features and function in biological system.	5
2	188.	What do you mean by ferridoxins? Give the examples of different types of ferredoxins and discuss their structural features.	5

2	189.	What is cytochrome c oxidase? Discuss the structural properties of it? How does it function.	5
2	190.	Discuss the structure and function of cytochrome c.	5
2	191.	Discuss the structure and activity of carboxypeptidase A(CPA).	5
2	192.	What is carbonic anhydrase? Discuss the role of zn(II) in carbonic anhydras.	5
2	193.	Explain the structure and reactivity of carbonic anhydrase.	5
2	194.	Explain the structure and reactivity of alcohol dehydrogenase enzyme?	5
2	195.	Give an account of the structure and reactivity of cytochrome P 450.	5
2	196.	Explain the structure of cytochrome P-450 and its catalytic cycle of cytochrome P-450 in the oxygenation reaction.	5
2	197.	Compare the mechanism of activity of catalase and peroxidase ?	5
2	198.	Discuss the structural features of Cu, Zn-SOD. Comment on the structure forming role of the Zn^{2+} in Cu,Zn-SOD. Discuss the mechanism of Cu, Zn-SOD.	5
2	199.	Discuss the structural properties of catalase and peroxidase? How do they function?	5
2	200.	Discuss the mechanism of function of cytochrome P-450. How do you explain the spin state change of Fe- centre in the enzyme with the attachment of the substrate?	5
2	201.	Discuss the structure and mechanism of the action of xanthin oxidase enzyme. How does the drug, allopurinol, function against gout?	5
2	202.	What is sulfite oxidase? Discuss its mechanism of activity.	5
2	203.	What is aldehyde oxidase? How does the chelating drug antabuse, work against alcoholism.	5
2	204.	Write a brief note on molybdenum containing enzymes.	5
3	205.	Explain the important and therapeutic uses of metal in medical field.	5
3	206.	Describe the role of metals and it complex compound found in plant and animals.	5
3	207.	Write a note on (a) Fe metal complex in human body, (b) Mg metal complex in plant body.	5

3	208.	Write a note on anemia.	5
3	209.	Explain briefly, what are the causes, symptoms and treatment of anemia disease?	5
3	210.	Write a note on deficiency of zinc in human body.	5
3	211.	Write a note on deficiency of following in human body (a) Iron (Fe) (b) Zinc (Zn)	5
3	212.	Explain the deficiency and treatment of copper in human body.	5
3	213.	Write a note on deficiency of copper.	5
3	214.	Write a note on "Keshan Disease".	5
3	215.	Write a note on (a) Anemia, (b) Keshan Disease	5
3	216.	Write the causes, symptoms and treatment of hyponatremia?	5
3	217.	Explain the diseases caused due to deficiency of sodium metal in the human body.	5
3	218.	Write a normal blood sodium level in human body and also write a note on disease caused due to deficiency of sodium in human body.	5
3	219.	Write the causes, symptoms and treatment for disease due to the deficiency of potassium in human blood.	5
3	220.	Write the causes, symptoms and treatment for hypomagnesaemia.	5
3	221.	Write a note on deficiency of calcium in human with its tratment.	5
3	222.	Explain the mode of action of metal complex in antibacterial agent.	5
3	223.	Write a note on metal complex used as antimicrobial agent.	5
3	224.	Write a note on Isoniazid as antitubular drug	5
3	225.	Explain the metal Complex which shows both antifungal and antibacterial activity	5
3	226.	Describe the property of Bleomycin in presence of Fe (II)?	5
3	227.	Write a metal complex and Chelating ligand used as (i) Antibacterial agent (ii) Antifungal agent (iii) Antiviral agent (iv) Antitubercular agent (v) Anticancer agent	5
3	228.	Explain the mode of action gold metal complex in treatment of rheumatoid arthritis.	5

3	229.	Explain the mode of action of platinum complex in cancer treatment.	5
3	230.	Write a note on metal complex used as (i) Anticancer drug. (ii) Anti rheumatoid drug.	5
3	231.	Explain the mode of action of vanadium complex in treatment of diabetes.	5
3	232.	Write a note on vanadium metal complex as anti diabetic agent?	5
3	233.	Discuss the mode of action of cyanide on human body?	5
3	234.	Explain the action and antidote of cyanide?	5
3	235.	Describe the process of histo-hypoxia.	5
3	236.	Describe the mechanism of lead ion toxicity.	5
3	237.	Explain the mechanism of antidote complex with poison like arsenic.	5
3	238.	Report the antidote accelerated metabolic conversion of poison like cyanide to non toxic product.	5
3	239.	Discuss the antidote used for conversion of poison like carbon monoxide to non toxic product.	5
3	240.	Write a note on mechanism of antidote complex with poison like mercury and copper.	5
3	241.	Review and report on mechanism of antidote complex with poison like iron (Fe) and copper (Cu).	5
3	242.	Write a note on metal complex used in radio diagnostic	5
3	243.	Explain the general aspect mechanism of metal ion toxicity.	5

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Question Paper Pattern- Model Question Paper

Q.P Code: 14MSCDH120

St. Philomena's College (Autonomous) Mysore

IV Semester M.Sc. Final Examination April/May 2018

Subject: CHEMISTRY

Time: 3 Hours **Title: BIO-INORGANIC CHEMISTRY- Hard Core**

Max Marks: 70

PART –A

1. **Answer the following:** **10x2=20**

a. Discuss the properties of iron which make it so important biologically.

b. Enumerate the characteristic features of B_{12} coenzyme.

c. What are the different conformations of DNA?

d. Write the structures of gold complexes which are used as drugs for arthritis.

e. Explain the mitochondrial electron transfer chain.

f. What do you mean by active transport and passive transport with examples?

g. What are the basic requirements of a dioxygen complex to function as an oxygen carrier?

h. Illustrate the phenomenon, metal-metal detoxification.

i. What are siderophores? Explain why high-spin Fe^{3+} complexes of siderophores are kinetically labile whereas analogous model complexes containing Cr^{3+} are kinetically inert.

j. What is Na+ pump? How does it function?

PART –B

Answer any five of the following: **5x10=50**

2 a Explain how Ca^{2+} ion makes conformational change after binding the active site of Troponin-C and explain how this alteration regulates muscle contraction.

b Give the coordination environment of cobalt in vitamin B_{12} coenzyme. Explain any two important biochemical reactions catalyzed by it. **6+4=10**

3 a What are the disease caused by the deficiency of iron, Zinc and copper? How are they treated?

b Explain the structural features and mechanism of activity of Cu-Zn SOD **5+5=10**

PTO

4 a What are the different blocking agents in the respiratory chain?

b Discuss the mechanism of drug action of cisplatin as an anticancer drug. Suggest the possible ways to mitigate side effects caused by the drug. 4+6=10

5 a Explain the structure of cytochrome P-450 and its catalytic cycle of cytochrome P-450 in the oxygenation reaction.

b What is Na^+-K^+ pump? How does it function? How do you explain their ion selectivity? 5+5=10

6 a. Explain with a schematic view the tetrameric structure of the hemoglobin.

b. Describe the intrinsic pathway of blood clotting process. 6+4=10

7 a. What are the symptoms of Arsenic-poisoning? Name the chelating antidotes used in Arsenic-poisoning. Comment on the possible role of Arsenic in the life process.

b Discuss the structure and biological functions of ferritin. 5+5=10

8 a Graphically represent the effect of essential metal and toxic metal on biological growth.

b What are the characteristic features of primary and secondary structure of proteins? 5+5=10

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